Equation of a tangent from a graph

1

a

i $(1, 0)$ ii 2 iii $y = 2x - 2$

b

i $(1, -3)$ ii -2 iii $y = -2x - 1$

c

i $(2, 2)$ ii -1 iii $y = -x + 4$

d

i $(4, 0)$ ii 1 iii $y = x - 4$

e

i $(2, 3)$ ii $\frac{1}{2}$ iii $y = \frac{x}{2} + 2$

2

a

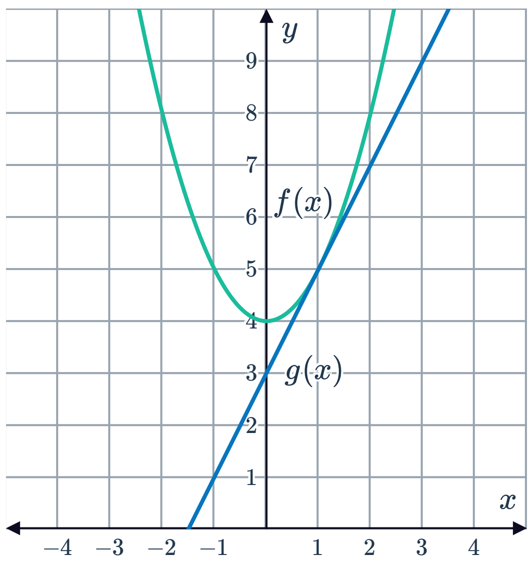
i $(-1, -6)$ ii 3 iii $y = 3x - 3$ iv $x = 1$

b

i $(1, -4)$ ii -3 iii $y = -3x - 1$ iv $x = -1$

3

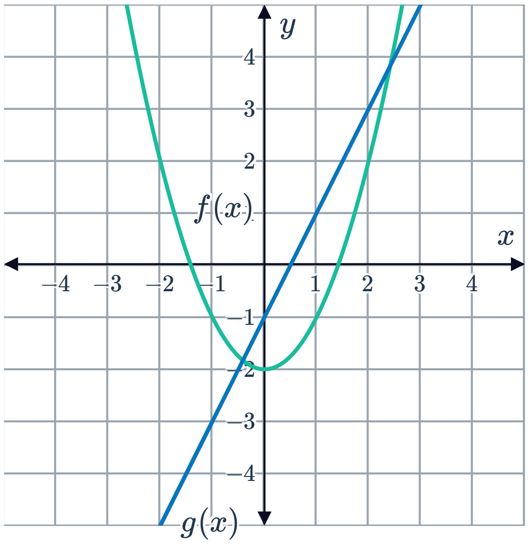
a



b Yes, because the line $g(x)$ touches the curve $f(x)$.

4

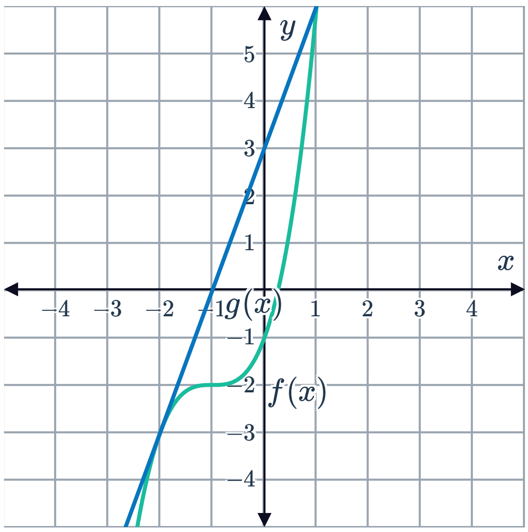
a



b No, because the line $g(x)$ crosses the curve $f(x)$ twice.

5

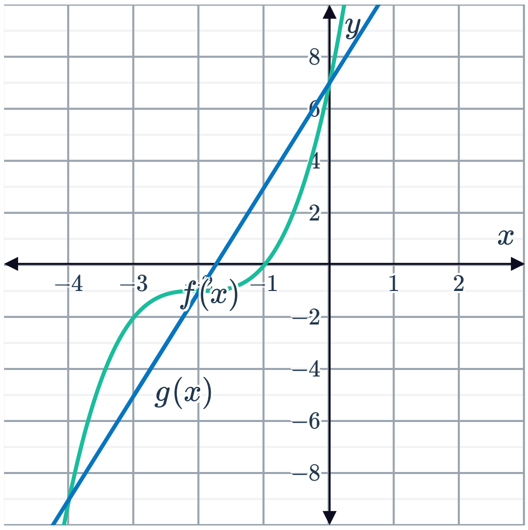
a



b Yes, because the line $g(x)$ touches the curve $f(x)$ at $x = -2$.

6

a

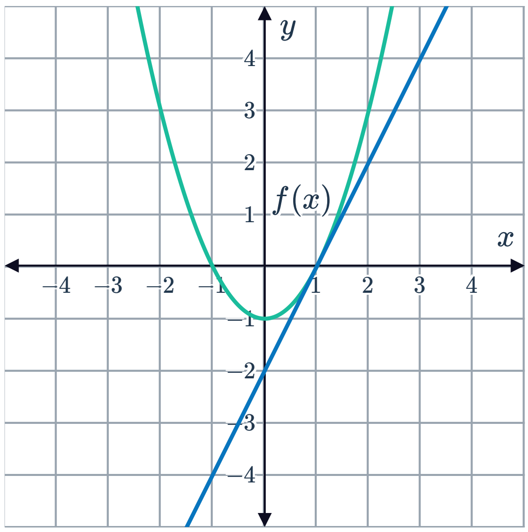


b No, because the line $g(x)$ crosses the curve $f(x)$ three times

7

a 2

b

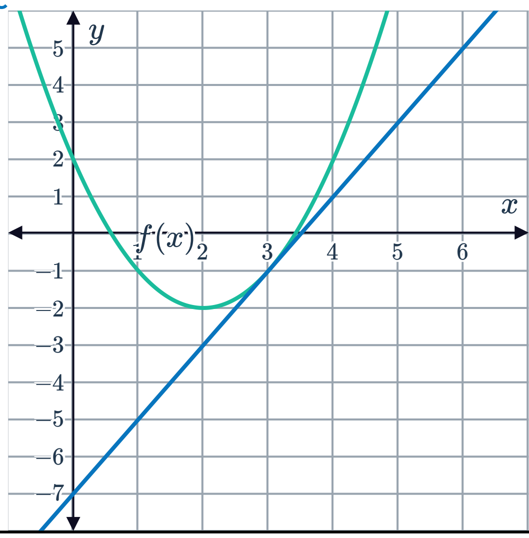


8

a 2

b $(2, -2)$

c



Equation of a tangent at a given point

9 $x = 2$

10

a $f'(-1) = -2$

b $y = -2x - 1$

11

a $f'(-1) = 3$

b $y = 3x + 2$

12

a $f'(-3) = 6$

b $y = 6x + 9$

13

a $f'(-2) = -4$

b $y = -4x - 4$

14

a $f'(2) = 12$

b $y = 12$

c $y = 12x - 12$

15 $y = -10.1x + 8.4$

Applications

16

a

i $x = 2$

ii $y = 4$

b

i $x = 2$

ii $y = 8$

17 $x = \frac{3}{4}, x = -2$

18

a $\frac{dy}{dx} = 2x + b$

b -5

c $b = -23$

d $c = 79$

19

a $x = 4$

b $(4, 36)$

20 **a** $x = 15, -11$

b $(15, 2025), (-11, -2057)$

21

a $x = 2, -4$

b $(2, -16), (-4, 62)$

22

a $x = 2, x = -6$

b $(2, 2), (-6, 82)$

23

a $\frac{dy}{dx} = 8x - 5$

b $x = \frac{5}{8}$

24 $a = 3, b = 5$

25 $a = 2, b = 3, c = 0, d = -28$